

Identifying Cardiovascular Disease Risk with Machine Learning

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Abstract—The name "cardiovascular disease" (CVD) called to a group of disease that affect the circulatory system, which includes the heart and blood vessels that carry blood throughout the body. This phrase refers to a group of conditions that collectively impair the cardiovascular system's ability to operate normally. CVD is the reason for 30% of all the world's deaths. Predicting CVD in a person is tough and very confusing. To improve the prediction accuracy, the authors have used the publicly available dataset of 304 people to train the model. The authors have applied four machine learning classifiers: K nearest neighbor, Naive Bayes, Logistic regression, and Decision tree. The authors applied two kind of feature selections LDA and SVA. Naive Bayes achieved the highest accuracy of 88.50% to detect if a person is suffering from CVD or not.

Keywords— CVD Detection, KNN, Logistic regression, Decision Tree, Naïve bayes.

I. INTRODUCTION

Cardiovascular disease (CVD) introduced as the principal global cause of mortality. This category encompasses various disorders affecting the cardiac muscle and the vascular system supplying crucial organs such as the heart and brain. In the early 20th century, CVD contributed to less than 10 percent of global deaths, but the scenario underwent a drastic shift, with the figure reaching 30 percent by 2001. Notably, approximately 80 percent of CVD-related deaths transpire in low- and middle-income countries. CVD would emerge as the leading cause of global mortality and disability by 2020, primarily driven by its escalation in low- and middle-income nations. Indeed, by 2001, CVD had assumed the leading cause of death in the developing world, mirroring its long-standing status in the developed world since the mid-1900s [1]. Detecting CVD is considered a tough task, but by detecting it in the early stage we can prevent death by a significant number. In England, around 62 million individuals are affected by CVD, while another 50 million are

dealing with hypertension. CVD carry on the No. 1 global cause of death, add up for 17.3 million deaths per year, this number that is expected to reach to 23.6 million by 2030 [2]. In recent years, the occurrence of traditional risk factors for atherosclerotic cardiovascular diseases has surged in populous developing nations like China and India. This rise has led to an increase in coronary and cerebrovascular events[3].

Early diagnosis plays a crucial role in providing timely and effective care for patients, impacting both individual health and the financial aspects of healthcare systems[4]. However, current prediction strategies, especially in CVD, are often inefficient and time-consuming. Processing large amounts of information about a patient's heart is even tedious for people working in the health sector. Using machine learning to create a model which provides accurate and early detection will be very helpful in this scenario.

Machine learning is already successful in the diagnosis of breast cancer, skin cancer, and other disease so it would be in cardiovascular disease

II. RELATED WORK

Researchers and scholar who have done similar work and provided their information on CVD detection are mentioned below.

Ramakrishnan [5] used the CHDD and applied several machine learning classifiers. The author used random forest (RF), decision tree classifier (DT), multilayer perceptron (MP), and XGBoost (XGB) and achieved the highest accuracy of 87.28%.

Reddy et al. [6] used CHDD. In this research, ten machine learning (ML) classifiers from different categories, such as

Lazy, meta, bayes, functions, rules, and trees by the author and achieved the highest accuracy of 86.4%.

In the study conducted by Alotaibi et al. [7], the researchers employed the Classification of CHDD in their investigation. Notably, they incorporated various techniques, including dimensionality reduction, Chi-squared, and NB, in their analytical framework. As a result of their comprehensive approach, the authors achieved a commendable accuracy rate of 85.15%.

Youna [8] used the CHDD. The authors applied 6 classifier techniques which are DT, SVM, KNN, NB, LR, and RF. The author achieved the highest accuracy of 86.89% using LR as compared to other classifiers.

Rani et al. [9] used the CHDD. The authors applied classifiers like Genetic Algorithm (GA) and recursive feature elimination and achieved the accuracy of 86.6%

Fahim et al. [10] used the CHDD. The authors applied A feature selection algorithm, which combines the GA with recursive feature elimination, was employed in a hybridized approach. This method aimed to identify and select optimal features from the given dataset like KNN, RF, DTs, and XGBoost and achieved the highest accuracy of 73.72%.

Bataineh et al. [11] used CHDD . The authors applied In this research, the study incorporates a hybrid algorithm known as MLP-PSO (Multi-Layer Perceptron-Particle Swarm Optimization). Alongside this novel hybrid algorithm, the investigation employs ten diverse ML algorithms. The collective use of these algorithms forms the basis of the study's predictive models for heart disease and achieved the highest accuracy of 84.61%.

Reddy et al. [12] used the CHDD. This evaluation was conducted both in the absence and presence of Principal Component Analysis (PCA), offering a comprehensive examination of the algorithms' performance under different conditions. The authors applied classifier like DT, DA, LR, NB and many more and achieved the highest accuracy of 85.8%

III. METHODOLOGY

A. Dataset

The authors performed the experiment on the UCI Cleveland Clinic Heart Disease dataset[13]. The dataset contains data of 303 patients (207 male and 96 female). There are a total of 76 attributes out of which 13 are selected as prediction and one is a result attribute. The result attribute categorizes a person as dead or alive.

In the dataset of 303, 176 patients (Target 1) have CVD, and 127 don't have (Target 0). Figure 1 shows a graphical representation of the number of patients who have CVD and patients who don't. Table 1 shows the total number of attributes.

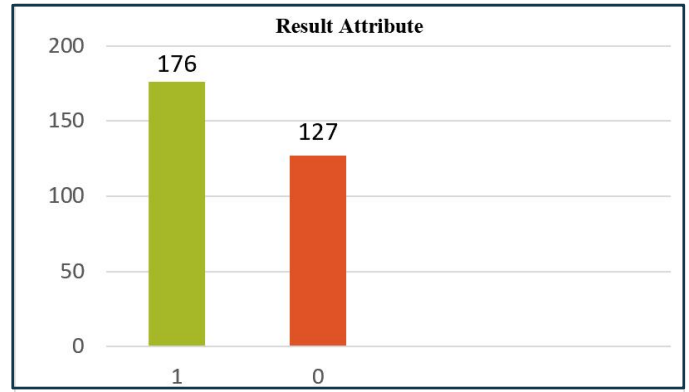


FIG. 1. BAR GRAPH REPRESENTATION OF RESULT ATTRIBUTE

TABLE I. TOTAL NUMBER OF ATTRIBUTES

SR. NO	FEATURES
1	Age
2	Sex
3	Chest pain
4	Rest blood pressure
5	Cholesterol
6	Fasting blood sugar
7	Resting electrocardiographic results
8	Highest heart rate
9	Exercise induced angina
10	Old peak
11	Peak workout ST segment
12	Count of major vessels
13	Nuclear stress examination
14	Target

Attribute Description :

- **Age** = In years of particular attribute
- **Sex** = If the patient is male then it is denoted by 1 in data and if the patient is female it is denoted by 0
- **Chest pain** = Chest pain is very common in CVD. The pain can range from discomfort to severe pain, which can be life-threatening
- **Rest blood** = Rest blood pressure is the blood pressure when the body is not under any physical stress. The range is less than 120mm Hg systolic and under 80mm Hg diastolic.
- **Cholesterol** = High level of cholesterol in blood can cause major heart problem
- **Fasting blood sugar** = Ideal fasting blood sugar level should be between 90-130mg/dL
- **Resting electrocardiographic results** =Resting ECG is when the patient is not moving. Resting heart rate should be between 60-100bpm
- **highest heart rate** = Highest heart rate is the maximum heart rate. It is the highest number of beats per minute that the heart achieved during Exercise.

- **Workout induced angina** = It is a type of chest pain that occurs when blood does not receive enough oxygen
- **Old peak** = a measure of abnormality in electrocardiograms. It is measured in Millimeter.
- **peak exercise ST segment** = The peak exercise ST segment is a part of the ECG reaction that is evaluated during an exercise stress test
- **Ca** = A feature used to identify heart disease. If ca equal to 0 the patient have heart disease.
- **Thal** = a diagnostic imaging test that measures the blood flow to your heart while you're resting and active
- **Target** = If the output is 1 the patient has CVD and if output is 0 patient don't have CVD.

B. Research Methodology

In their study, the researchers employed four classifiers (NB, LR, DT, and KNN) and two feature selection techniques (LDA and SVD). The research methodology, outlined in Figure 2, guided the experiments. The dataset was divided into test and training sets, and models were trained using these data. The accuracy of the models was then assessed to determine whether individuals had heart disease or not. This concise approach highlights the researcher's comprehensive exploration of heart disease prediction using various classifiers and feature selection methods.

C. Feature Selections

Singular Value Decomposition (SVD) is an important matrix factorizations of the programmatic era, It provide a numerical balanced matrix decomposition that can be utilised for many objective. It also exercise as the underlying algorithm of Principal component analysis(PCA) [14] .

Linear Discriminant Analysis (LDA) is a very often used solution for dimensionality reduction related issues as an initial processing step for ML. It increase the percentage of the between class inequality to the within class variance, thus confirming maximum class separation [15].

D. Classifier's

KNN: K-Nearest Neighbor (KNN) classifier achieved the classification by finding the closest neighbours to a line and benefitting by those neighbours to identify the group of a query. This method of classification is mainly important because the weak runtime result is not a big deal these days due to computational power[16] .

This technique mostly useful when we need to take more than one neighbour in our account so the method is called K-Nearest Neighbour classification. KNN neighbour are utilised to identify the class.

NB: The NB classifier uses Bayes' rule together with a hard believe that the features are conditionally independent given the class. Even though this independence believe is many times proved wrong in practice, NB despite everything often delivers competitive classification accuracy. NB gives a system for utilising the information in sample y data to find the posterior probability $p(y | x)$ of every class y given and object x. Joined

with its logical efficiency and many more desirable feature, it shows NB widely being in practice[17].

LR: Logistic Regression was introduced in late 1960 and beginning of 1970 as an alternative of ordinary least square (OLS). After that time LR utilisation increased in social science and in ai research. Generally LR is widely used for describing and testing hypotheses about between a categorical outcome and continuous predict variable[18].

DT: Decision Tree is the most famous method for representing classifier. Nodes are available in DT which form a rooted tree, The directed tree with a node called "root" that has zero returning edges. All remaining nodes have exactly one incoming edge. Outgoing edges node is called internal or test node. All remaining nodes are called leaves(also called as terminal or decision node). Whenever numeric attribute, decision trees can be geometrically interpreted as a collector of hyperplanes, each orthogonal to one the axes. The goal is to find the optimal decision tree by minimizing the generalization error[19].

E. Accuracy

Accuracy is calculated by dividing the number of correct predictions by the total number of predictions made.

$$\text{Accuracy} = \frac{\text{Total number of correct prediction}}{\text{Total Number of Prediction}} \times 100$$

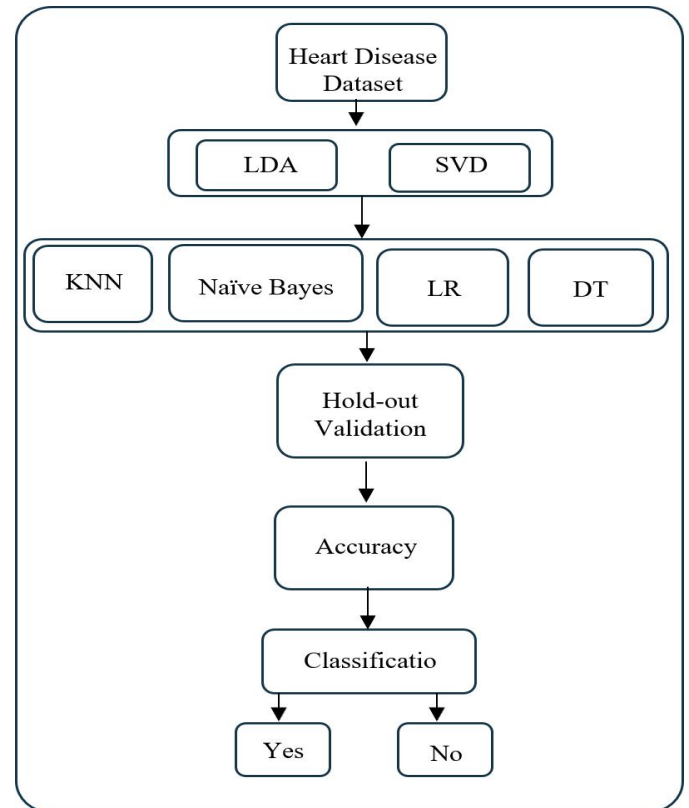


FIG 2. RESEARCH METHODOLOGY

IV. RESULT AND METHODOLOGY

The researcher used four different classifiers and applied different kinds of filtration methods to the CVD dataset. The classifier's performance was compared without filtration and with filtration and the highest accuracy achieved is 88.50% by NB. The other classifiers are KNN, LR, and DT. Their accuracy without filtration is 62%, 81%, and 65%. Their best accuracy with filtration is 86%, 85%, and 77%. Table 3 shows the difference between related work accuracy and our accuracy.

Figure 2 shows graphical representation of accuracy without feature selection. Figure 3 shows graphical representation of accuracy with SVD feature selection. Figure 4 shows graphical representation of accuracy with LDA feature selection. Table 2 shows performance of classifier with and without different filtration method. The accuracy comparison of each classifier shown in Figure 5. Figure 6 represents graph representation of accuracy label.

TABLE 2. CLASSIFIER'S PERFORMANCE

Features Classifiers	Without Feature	Single value Decomposition	Linear discrimination Analysis
KNN	62%	85%	86%
NB	81%	88.5%	82%
LR	81%	85%	83%
DT	65%	75%	77%

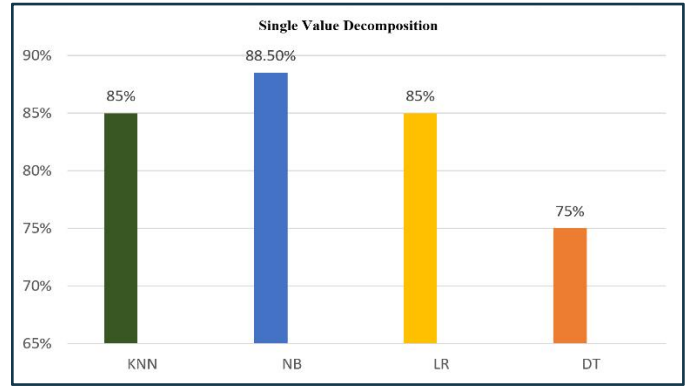


FIG 3. ACCURACY COMPARISON USING SVD

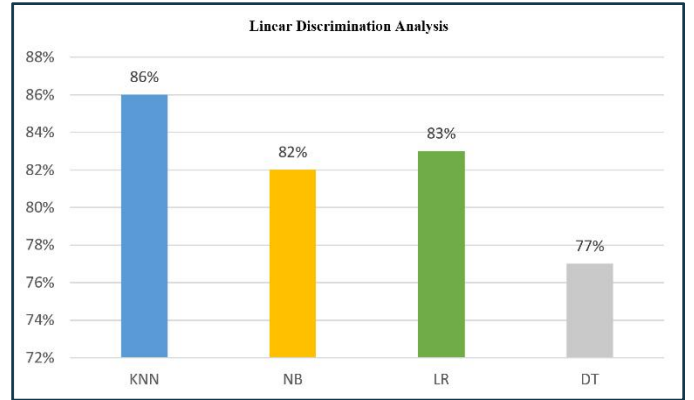


FIG 4. ACCURACY COMPARISON USING LDA

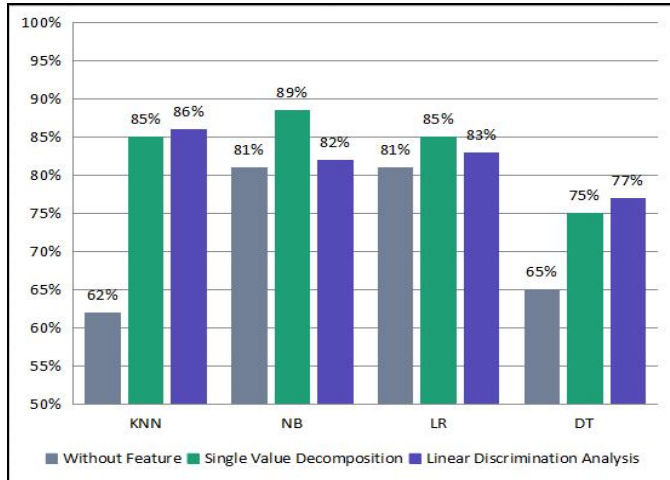


FIG 6. ACCURACY COMPARISON

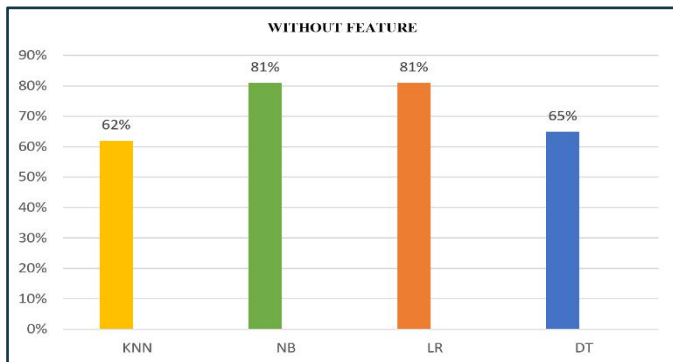


FIG 2. ACCURACY COMPARISON WITHOUT FEATURE SELECTION

FIG. 5 COMPARISON OF RELATED WORK

Authors	Year	Dataset	Accuracy
Ramkrishnan	2024	UCI	87.28%
Reddy et al.	2024	UCI	86.4%
Alotaibi et al.	2025	UCI	85.15%
BABAJIDE	2024	UCI	81.9%
Rani et al.	2025	UCI	86.6%
Bataineh et al.	2023	UCI	84.61%
Reddy et al.	2026	UCI	85.8%
Fahim et al.	2025	UCI	73.72%
	2026	UCI	88.50%

V. CONCLUSION

In the quest to predict CVD, the researcher proposed employing a machine learning model featuring four classifiers: LR, NB, DT, and KNN. Notably, NB emerged as the top performer, achieving an impressive accuracy of 88%. It outperformed all other classifiers in the evaluated performance parameters.

The high accuracy and robust performance of Naive Bayes make it a valuable tool for clinicians seeking to predict CVD with enhanced success rates at an early stage. The study also suggests avenues for future research, encouraging further experiments with ensemble classification methods to potentially elevate the accuracy of the existing model. This paves the way for continual refinement and improvement in predicting heart failure through advanced machine learning approaches. The high accuracy and robust performance of NB make it a valuable tool for clinicians seeking to predict CVD with enhanced success rates at an early stage. The study also suggests avenues for future research, encouraging further experiments with ensemble classification methods to potentially elevate the accuracy of the existing model. This paves the way for continual

refinement and improvement in predicting heart failure through advanced machine-learning approaches

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